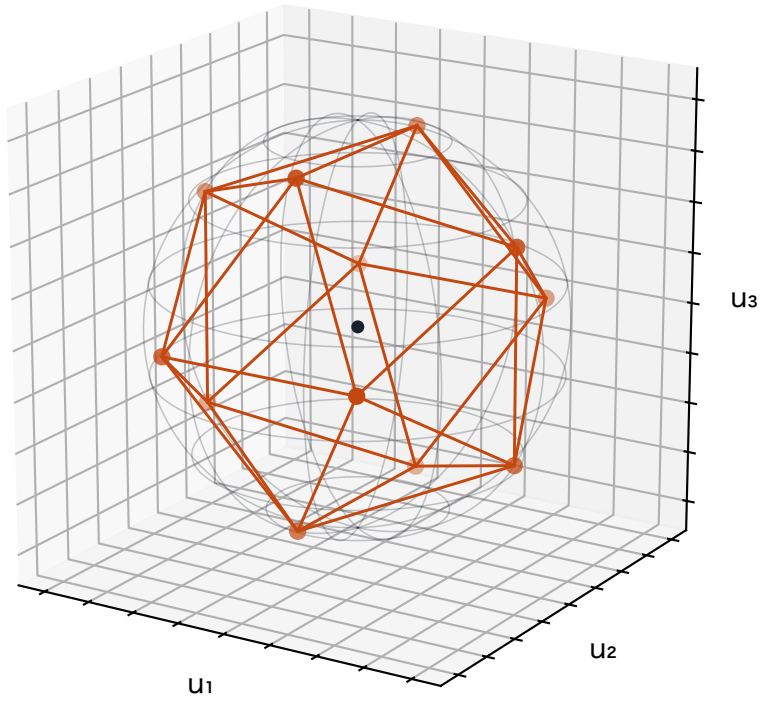
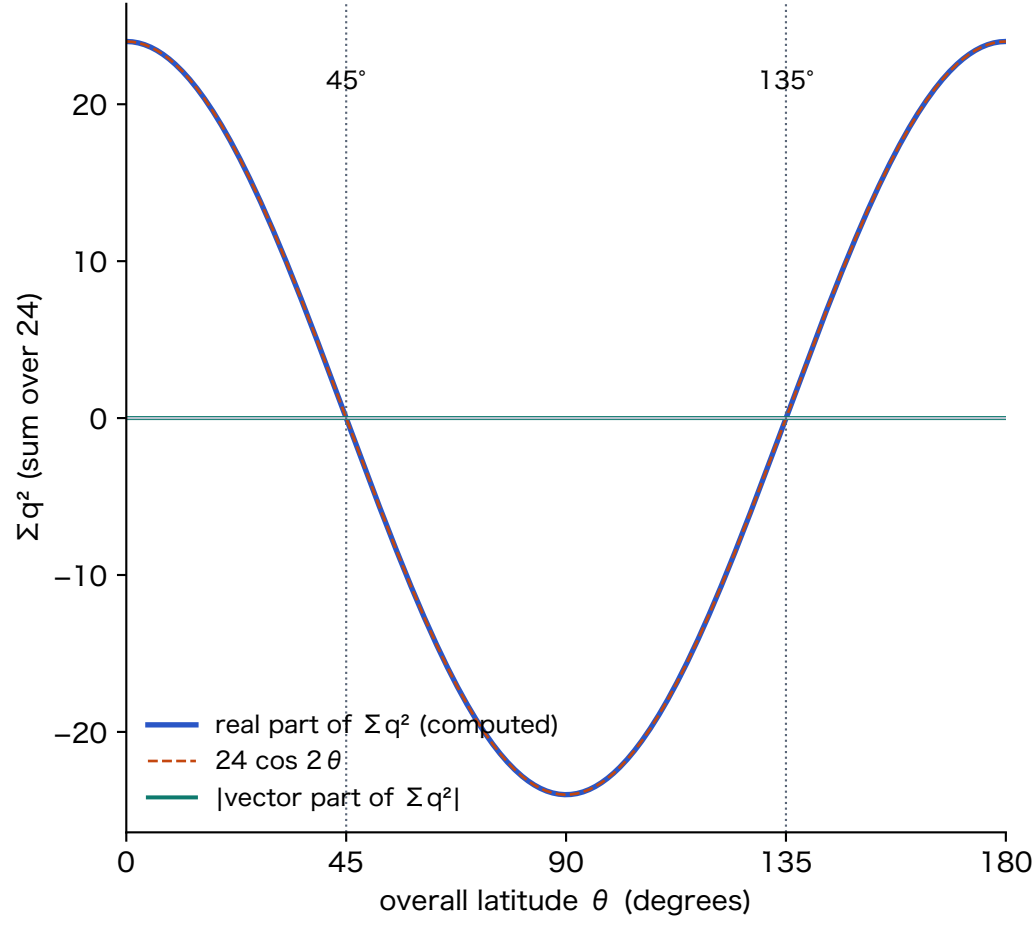
Figure 6. The most symmetric form q = ±e^(π n/4) = ±(1 + n)/√2 (n the 12 icosahedral directions, N = 24)

image u = v/q₀ = n in the tangent hyperplane q₀=1 (24 q map to 12 points).
All lie on |u|=1, the image of the light cone; edges are straight (icosahedron)



For the 24 quaternions q = ±(cos θ + n sin θ), Σq² = 24 cos 2θ.
It vanishes only at θ=45° (q = ±e^(π n/4)) and 135°